

HARRY WU

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EXPERIENCE

Keep Your Head Up Foundation

May 2025 – Aug. 2025

Website Developer / SEO & Automation Engineer

Waterloo, ON

- Improved Google ranking for “Waterloo concussion” from **page 3+** to **page 1 (top 5)** by optimizing metadata, headings, internal linking, and indexing
- Standardized URLs and headings across **30+ pages** to improve crawlability
- Integrated forms, CMS, analytics, and automation via APIs; added validation/logging to reduce failures
- Built **3+** Zapier workflows linking forms and Sheets; reduced manual data handling by **~40%**
- Tracked impressions, CTR, and indexing weekly in Google Analytics and Search Console to guide SEO iterations
- Added unit tests and regression checks for key scripts to keep outputs stable during frequent updates

PROJECTS

SEO Website Generator & Optimizer | JavaScript, HTML/CSS, Node.js, OpenAI API

Nov. 2025 – Jan. 2026

- Built a generator that turns business context, location, and keyword intent into SEO-ready webpages
- Designed **3** reusable templates and structured prompts with output validation for consistent, deterministic HTML
- Implemented Git-based regenerate-and-diff on **20** inputs to compare outputs across prompt/template changes

AI-Enhanced Web Components Library | React, JavaScript, Python, Gemini API

Jul. 2025 – Jan. 2026

- Built a reusable library of **10+** React components with accessibility-first patterns and structured UI copy generation
- Built a REST service with **4** endpoints to generate consistent UI copy and component props for faster prototyping
- Standardized **5+** prompt templates and a JSON output schema to keep responses deterministic
- Wrote **20+** unit tests for component logic and API contracts to improve refactor reliability

Document Q&A RAG Prototype | Python, SQLite, OpenAI API + Embeddings

Jan. 2026 – Present

- Built a PDF Q&A pipeline that indexes documents in SQLite and retrieves top-ranked passages with citations
- Tuned chunk size/overlap across **4** configurations and refined prompting to improve retrieval relevance
- Built an eval harness with **5+** queries and expected citations to validate retrieval quality after changes

TECHNICAL SKILLS

AI & Automation: OpenAI/Gemini API + Embeddings, prompt engineering, structured prompting, RAG

Programming: Python, JavaScript, TypeScript, C++, Java, Bash, Git, unit testing

Full-Stack: React, Next.js, Node.js, Express, REST API, HTML, CSS

Databases: SQL, SQLite, PostgreSQL

SEO & Analytics: Technical SEO, Google Analytics, Google Search Console, keyword research, page optimization

Other Tools: Visual Studio, Zapier, Google Workspace, Microsoft 365, Google Sheets, Microsoft Excel

EDUCATION

Bachelor of Computer Science, President's Scholarship of Distinction

Waterloo, ON

University of Waterloo

Sep. 2024 – May 2029

- **Relevant Coursework:** Data Structures and Algorithms (C/C++), Object-Oriented Programming (C++)